



Pediatric Neuromusculoskeletal Disorders

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Recommended Reading

- Kliegman RM. *Nelson: Essentials of Pediatrics*. 7th Edition. Saunders; 2014
 - Chapters 179-187
 - Text available online via TUCOM library (Call# RJ45)

○ **Quizlet:** http://quizlet.com/_10ejsp =
Pediatric Neuromusculoskeletal Disorders

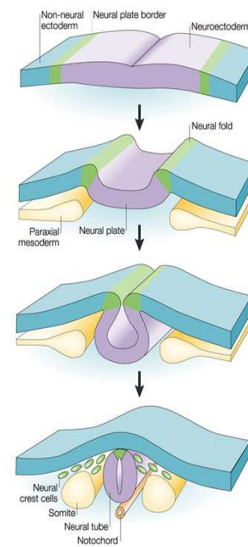
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Objectives

- Recognize the range of anomalies associated with spinal dysraphism (spina bifida occulta, meningocele & myelomeningocele) and their clinical manifestations
- Identify the major symptoms and common causes of increased intracranial pressure (ICP) (specifically, Pseudotumor Cerebri, Meningitis and Hydrocephalus)
- Recognize the specific clinical features of Spinal Muscular Atrophy, Duchenne's Musculodystrophy, Myotonic Dystrophy and Guillan-Barre Syndrome
- Distinguish between the presented neurocutaneous disorders (TS, NF1, NF2, SW) and recognize their clinical features

Development of the CNS

- Neural plate develops at **18 days** of gestation
- Gives rise to the **neural tube**
 - Forms the brain and spinal cord
- Also gives rise to the **neural crest cells**
 - Form the peripheral nervous system, meninges, melanocytes and adrenal medulla



Development of the CNS

Congenital Anomalies of the Nervous System

Neurologic Evaluation

Increased ICP

Psychomotor Retardation

Weakness

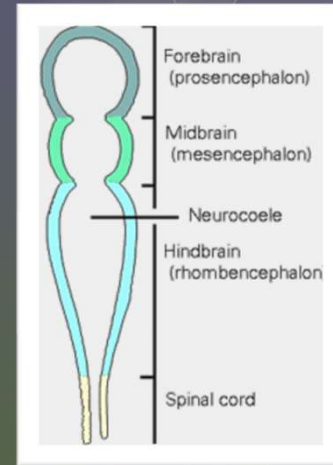
Muscle Disease

Neurocutaneous Disorders

Neural plate of the embryonic ectoderm is the precursor of the nervous system

Development of the CNS

- Neural tube begins to form at 22 days of gestation
 - **Rostral** end forms the **brain**
 - **Caudal** region forms the **spinal cord**
- Lumen of the neural tube forms the brain's ventricles and central canal of the spinal cord



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Congenital Anomalies of the Nervous System

○ **Spinal Dysraphism**/Neural Tube Defects

- Defective closure of the caudal neural tube

- Range of anomalies

- Defects of L5 or S1 vertebral arches

- **Spina bifida occulta** – defects of underlying bone or canal

- Sacral dimple, tuft of hair

- **Meningocele** – sac filled with CSF due to protrusion of the dura mater

- **Myelomeningocele** – spinal canal & meninges exposed

○ **Folate**

Development of the CNS

Congenital Anomalies of the Nervous System

Neurologic Evaluation

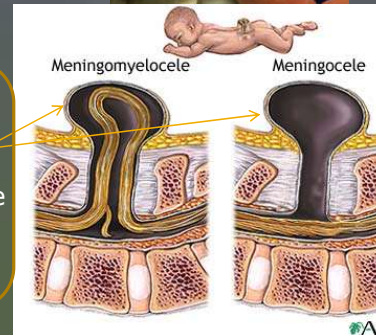
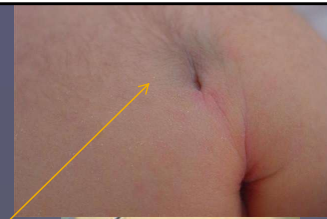
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End of the 4th week of gestation; myelocele = total paralysis and loss of sensation in legs, bowel/bladder incontinence; meningocele = spinal cord anatomically & functionally intact; spina bifida occulta – skin is intact – dimple, tuft of hair; Tethering of spinal cord to filum terminale, Lipoma or dermoid cyst, Tiny epithelial tract; may have difficulty with continence, numbness of feet, repeated meningitis

Folate is essential for normal development of the fetal nervous system, perhaps because of its ability to prevent elevated levels of homocysteine in amniotic fluid

Spinal Dysraphism/Neural Tube Defects



Preparing for Surgery on a Meningocele



Spina bifida occulta



Meningocele



Myelomeningocele

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Neurologic Evaluation

- Varies with age
- **Infant**
 - Examine the hair, skin, teeth & nails
 - Head circumference
 - Fontanel
 - Primitive reflexes
 - Posture
 - Movement and tone



Moro, grasp, rooting, trunk incurvation, placing, etc

Neurologic Evaluation

- Examine at **All Ages**:
 - Cranial Nerves
 - Motor Exam
 - Strength
 - Tone
 - Coordination
 - Gait
 - Reflexes
 - Involuntary Movements
 - Sensory Exam
 - Mental Status



Development of
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**Neurologic
Evaluation**

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Psychomotor
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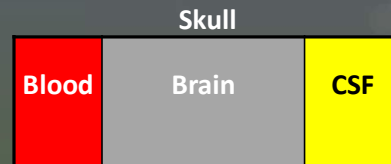
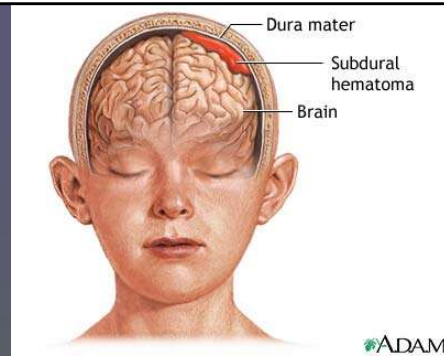
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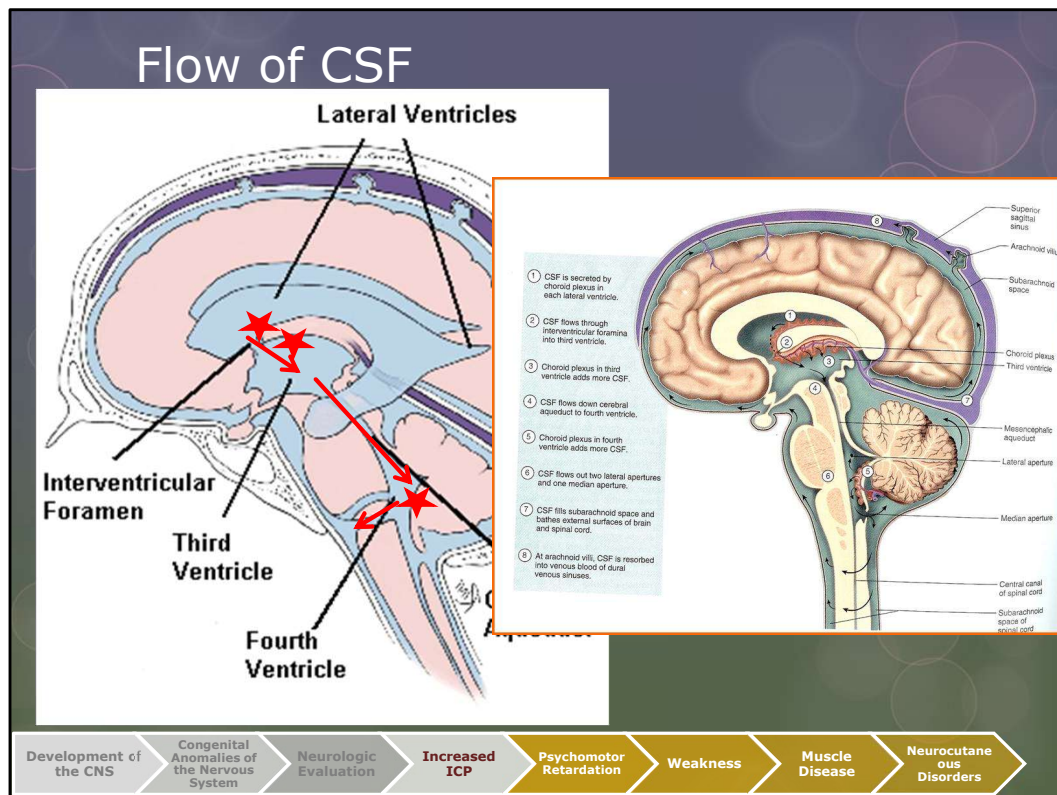
- Pathophysiology
 - Skull is a **rigid container**
 - **Brain, blood** and **CSF**
 - Mass lesion, hydrocephalus, cerebral edema, intracranial hemorrhage



Brain (80-85%), CSF (10-15%), Blood (5-10%)

Brain can accommodate raised ICP initially by expelling CSF & blood into the spinal subarachnoid space -> brain herniation

LP contraindicated in presence of increased ICP because leak of CSF promotes brain shifts and herniations



(lateral vent -> foramen of Monroe -> 3rd ventr -> cerebral aqueduct -> 4th ventr _> foramen of Magendie & Luschka)

Increased ICP – Signs & Symptoms

- **Headache** (morning), **vomiting** (morning), lethargy, irritability, 6th nerve palsy, strabismus, diplopia, **papilledema**
- **Bulging fontanel**, suture diastasis, distended scalp veins, sunsetting of eyes, **rapid growth of head circumference**
- Critically increased ICP (abnormal pupils)
 - **Cushing triad:**
 - **↑BP**
 - **↓ Pulse**
 - **Irregular Respirations**

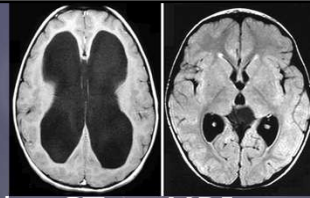


Cushing triad – late finding – elevated BP, decreased pulse, irregular respirations

Setting-sun sign refers to the downward deviation of an infant's or a young child's eyes as a result of pressure on cranial nerves III, IV, and VI.

Critically increased ICP = dilation & poor reactivity of one or both pupils

Increased ICP



- Diagnosis – HC, Sx&Sx, causes dx by CT or MRI

- Conditions with Focal Lesions on CT Scan

- Brain abscess, mass lesion

- **Hydrocephalus**

- Stretching of corticopontocerebellar & corticospinal pathways → **ataxia & spasticity**

- Distention of 3rd ventricle → compression of hypothalamic regions → **endocrine dysfunction**

- Distention of 3rd ventricle → compression of optic nerves, chiasm and tracts → **visual dysfunction**

- Tx = medical & surgical



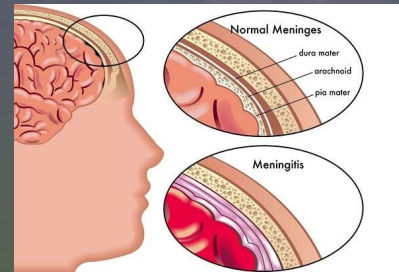
Abscess/mass – blockage of CSF, venous outflow or production of CSF

Hydrocephalus – slow, evolving syndrome over weeks/months – pressure from enlarging ventricles and interstitial edema in periventricular white matter – etiology = obstruction of CSF along its course (lateral vent -> foramen of Monroe -> 3rd ventr -> cerebral aqueduct -> 4th ventr -> foramen of Magendie & Luschka) or overproduction (choroid plexus papillomas)

Increased ICP

Conditions with Negative CT Scan & Abnormal MS

- Hypoxic-ischemic injury, trauma, infection, metabolic derangement or toxic ingestion
- **Meningitis**
 - Blockage of CSF pathways
 - Toxic cerebral edema
 - from hypoxic ischemic injury
 - Increased cerebral blood flow
 - Multifocal cerebral infarctions
 - Most can safely undergo LP
 - Contraindications = focal neurologic signs, poorly reactive pupils & tense fontanel



Abscess/mass – blockage of CSF, venous outflow or production of CSF – sx of infxn

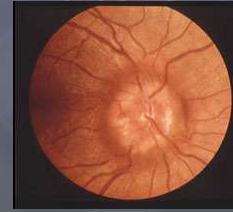
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Increased ICP

Conditions with Negative CT Scan & Normal MS

○ Pseudotumor Cerebri

- Benign
- **Obese female** of childbearing age
- Diffuse brain swelling vs CSF outflow resistance vs intracranial hypertension
- NOT critically ill
- **HA, diplopia**, abducens palsy & **papilledema**
- **Brain imaging = normal**
- Most often idiopathic, but can be certain drugs, endocrine disturbances, intracranial venous sinus thrombosis
- Tx = acetazolamide, repeated LPs, corticosteroids, surgery



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Neurologic Evaluation

Increased ICP

Psychomotor Retardation

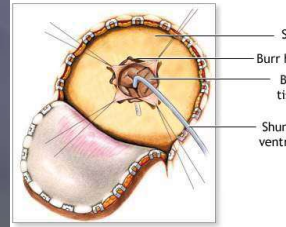
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Muscle Disease

Neurocutaneous Disorders

Increased ICP - Treatment

- Elevate head of bed (**reverse trendelenburg**)
- **Mannitol** – osmotic shift
- **Corticosteroids** – reduction of edema
- **Acetazolamide & furosemide** – transiently decrease production of CSF
- **Hyperventilation** – cerebral vasoconstriction & decreased cerebral blood volume
- **Ventricular catheter** – remove CSF & track ICP
- **Pentobarbital induced coma** – suppresses cerebral metabolism and blood flow



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Psychomotor Retardation Nonprogressive



- Very limited thought & physical movements
- **Static encephalopathy**: antenatal/perinatal disturbance or underlying genetic disease
- No neurologic focal deficits
- Normal lab exams
- Tx = infant stimulation & education programs

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Psychomotor Retardation **Progressive**

- Requires **complete laboratory & imaging evaluation**
 - Degenerative hereditary & metabolic diseases
 - Severe epilepsy
 - Chronic drug overdose (lead poisoning, vitamin deficiencies)
 - Hypothyroidism
 - Indolent brain infections (measles, syphilis)
 - Depression

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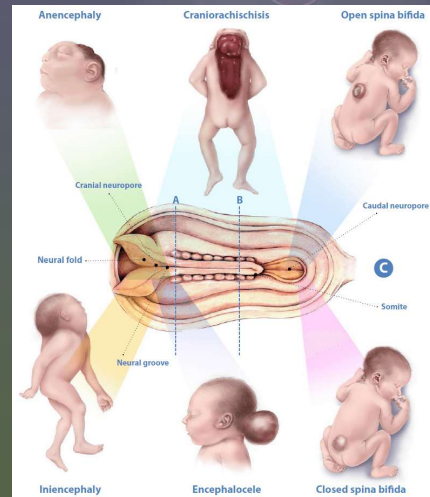
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Psychomotor Retardation – Congenital Malformations of the Brain

- Variety of injuries during a vulnerable period of gestation
 - Congenital hydrocephalus
 - Holoprosencephaly
 - Hydranencephaly
 - Schizencephaly
 - Lissencephaly
 - Macrogyria & polymicrogyria
 - Agenesis of the corpus callosum
 - Megalencephaly
 - Microcephaly



Chromosomal, metabolic, genetic, infxn, exposure to radiation, drugs, maternal illnesses

Congenital hydrocephalus - Congenital hydrocephalus is a buildup of excess CSF within the brain that is present at birth

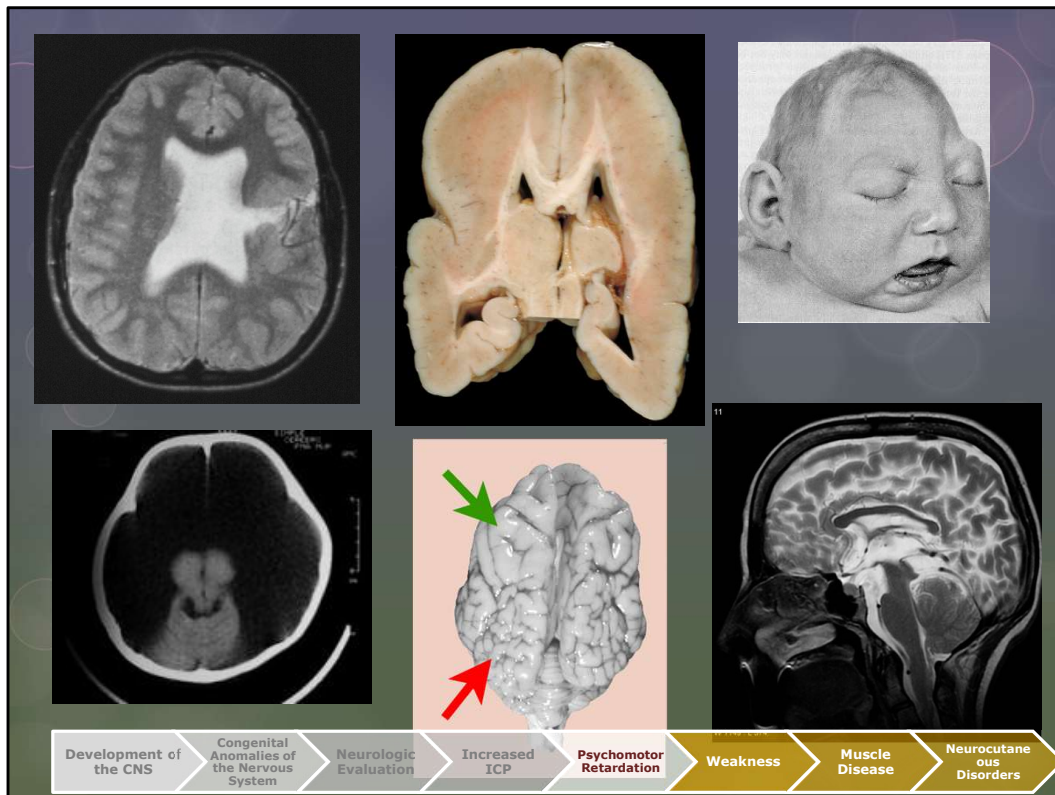
Hydranencephaly = brain develops normally until a likely vascular insult->absence of cerebrum with intact skull

Holoprosencephaly = varying degrees of failure of the primary cerebral vesicle to divide & expand laterally

Schizencephaly = abnormal slits, or clefts, in the cerebral hemispheres of the brain

Lissencephaly= "smooth brain," absence of normal convolutions in the cerebral cortex and an abnormally small head (microcephaly)

Megalencephaly= abnormally large, heavy, and usually malfunctioning brain



Schizencephaly, lissencephaly, microcephaly, hydranencephaly, polymicrogyria, agenesis of the corpus callosum

Weakness

- Upper Motor Neuron
- Lower Motor Neuron
- Neuromuscular junction
- Muscles



Weakness

- **Stroke** (Upper Motor Neuron)
- Werdnig-Hoffman or **Spinal Muscular Atrophy** (Lower Motor Neuron)
 - One of the most frequent autosomal recessive diseases (carrier = 1 in 50)
 - Progressive degeneration of the anterior horn cells
 - May begin intrauterine or anytime after
 - May progress rapidly or slowly
 - Earlier = rapid = worse
 - **Progressive proximal muscle weakness**
 - Flaccid quadriplegia, respiratory failure and death
 - Eyes remain bright and engaging



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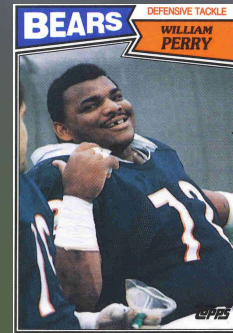
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Weakness – Guillain-Barre Syndrome

- Idiopathic peripheral neuropathy
- Often occurs after a respiratory or GI infection
- **Areflexia, flaccidity, symmetric ascending weakness**
- Resolves spontaneously



Campylobacter jejuni

Areflexia with normal muscle tone initially

In GBS, the immune system attacks the peripheral nervous system and starts to destroy the myelin sheath that surrounds the nerves.

Need tx in ICU

Weakness – Myasthenia Gravis

- Disorder of neuromuscular junction
- Classically begins in **teenage years**
- **Acute onset ptosis**, diplopia, ophthalmoplegia, weakness of extremities, neck and face
- Worsen as day progresses
- EMG has decremental response to repetitive nerve stimulation
- Anti-acetylcholine receptor antibodies



Some children only have eye findings, others require intubation

Tx = acetylcholine esterase inhibitor (pyridostigmine), possible intubation

Muscle Disease – **Duchenne's Musculodystrophy**

- Common **sex-linked recessive** trait
 - Incidence 20-30:100,000 (~1/3 = new mutations)
- Symptoms develop ~ 3 yrs – difficulty running
- **Calf pseudo-hypertrophy** (fat fibrosis)
- **Proximal leg weakness**
- Hyperlordotic & waddling gait
- **Gower Sign**
- Arm weakness ~ 6 yrs
- Wheelchair ~ 12 yrs
- Respiratory issues ~ 16 yrs



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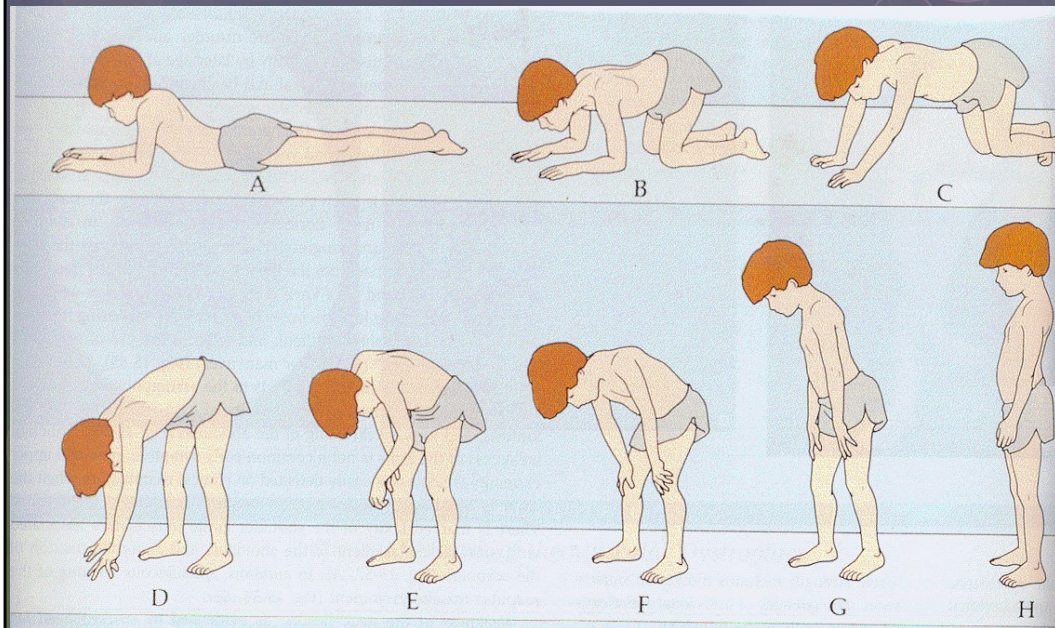
Weakness

**Muscle
Disease**

Neurocutane
ous
Disorders

Dystrophin is a protein associated with the muscle fiber plasma membrane – fiber degeneration & regeneration with increased intrafascicular connective tissue

Gower Sign



Muscle Disease – **Musculodystrophy**

- Serum **creatine phosphokinase** levels markedly elevated
- Due to absence of dystrophin
- **Becker muscular dystrophy** = abnormality in same gene locus leading to abnormal dystrophin
 - Later onset & slower progression



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Muscle Disease – Myotonic (Muscular) Dystrophy



- Autosomal dominant
 - 2 types
 - Gene designated Myotonin Protein Kinase
- Infant Form (type 1): **At birth = severe generalized weakness**
 - Expressionless faces
- Adult Form (type 2): In adolescence = **slowly progressive facial & distal extremity weakness**
- Characteristic facial appearance = hollowing of muscles, ptosis & lower lip drooping

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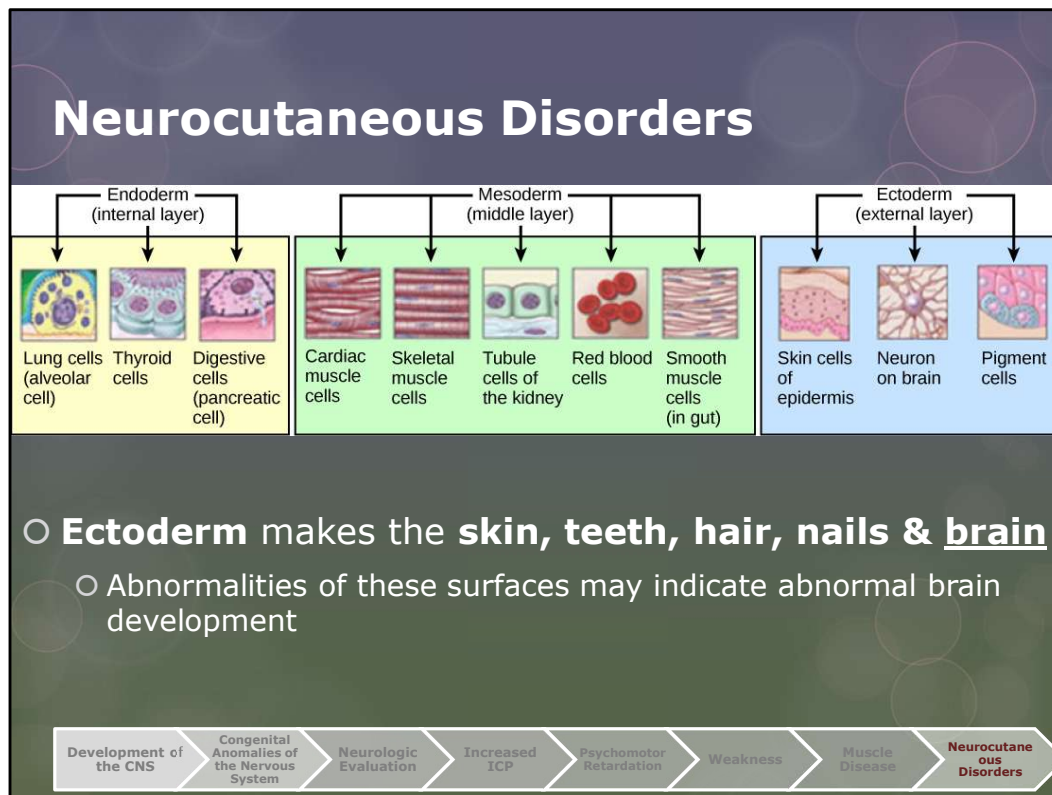
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Dystrophin is a protein associated with the muscle fiber plasma membrane – fiber degeneration & regeneration with increased intrafascicular connective tissue



Hypopigmented macules, renal angiomyolipoma, cardiac rhabdomyoma, brain tubers

A shagreen patch is a firm yellowish-red or pink area of nodules slightly elevated above the surrounding skin and often described as having the texture of an orange peel.

The ash-leaf spot is the earliest skin lesion in patients with tuberous sclerosis. This hypomelanotic macule is most easily visualized with a Wood's light. Ash-leaf spots are found at birth or in early infancy in 87 percent of patients with the disorder. By themselves, these hypomelanotic macules are not diagnostic of tuberous sclerosis because they are also relatively common in the general population

Ungual fibromas are flesh-colored papules that may displace and distort the nail bed.

Tuberous Sclerosis



- Autosomal dominant, incidence 1:10,000
- Hamartomas on organs: brain, eye, skin, kidneys & heart
- Facial angiofibromas, intellectual disability & severe epilepsy
- **Shagreen patches** (orange peel), **ungual fibromas**, **retinal hamartomas**, **ash-leaf spots** (hypopigmented macule)



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Tuberous Sclerosis



Shagreen patch, ash-leaf spot, retinal hamartomas, ungual fibromas

Neurocutaneous Disorders – **Neurofibromatosis 1**

- Autosomal dominant, incidence ~ 1:3,000
- Mutation of NF1 gene, chrom 17 q11.2
 - Tumor suppressor gene
 - Neurofibromin is a major negative regulator
 - New mutations ~30-50% of cases
- **Tumors grow in the nervous system**
- Most common = **neurofibromas**
 - Tumors of the peripheral nerves
- **Learning disability, scoliosis, macrocephaly, HA, optic gliomas**
- **Malignancy is the most common cause of death**
- Average life expectancy decreased by 10-15 yrs



Sphenoid wing dysplasia, cortical thinning of the long bones with pseudoarthrosis, peripheral nerve malignancy, pheochromocytoma, renovascular HTN, epilepsy

Café-au-lait spots = sharp borders, well demarcated, flat hyperpigmented lesions (“birthmarks”)

Neurocutaneous Disorders – Neurofibromatosis 1 Diagnosis



≥ 2 of the following:

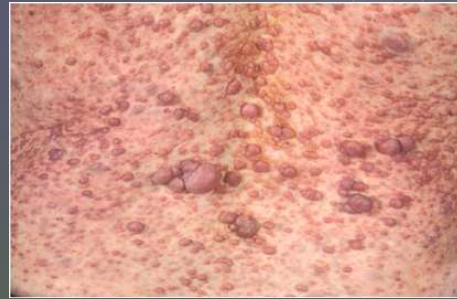
- ≥ 6 “café-au-lait” spots
 - >5 millimeters in diameter in children, >15 millimeters in adolescents and adults
- ≥ 2 neurofibromas
- Freckling in the area of the armpit or the groin
- ≥ 2 Lisch nodules or iris hamartomas
- An optic nerve glioma
- Scoliosis, abnormal development of the sphenoid bone, or the tibia
- A parent, sibling, or child with NF1

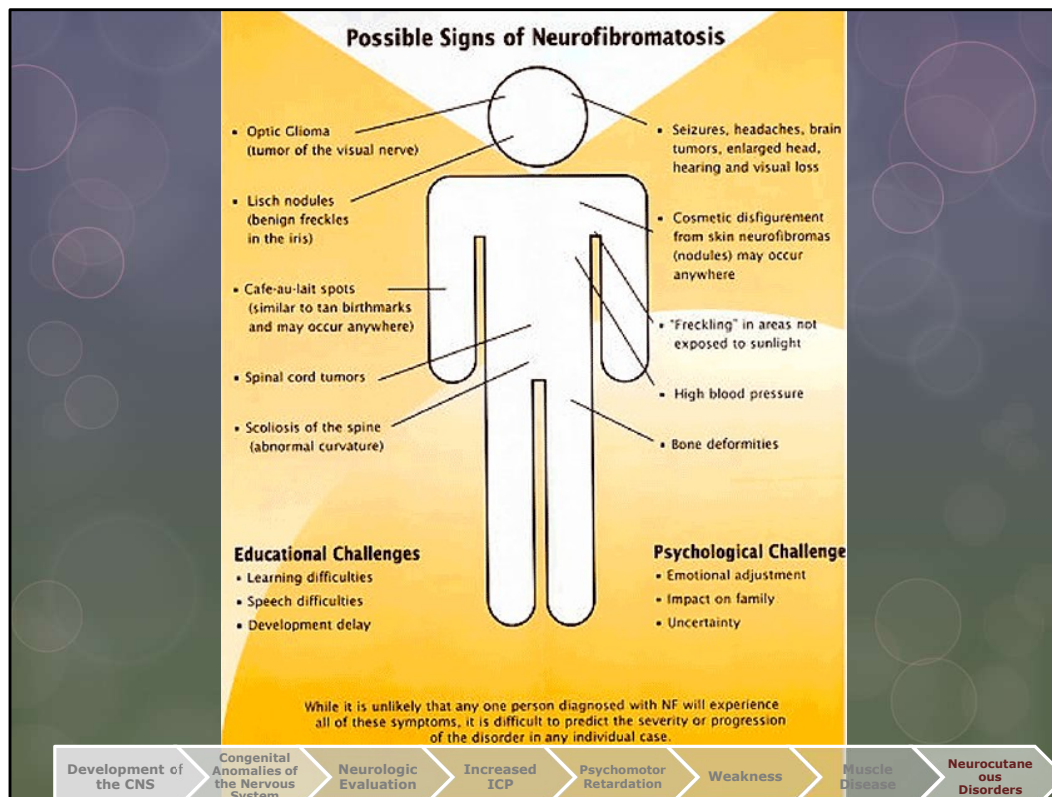


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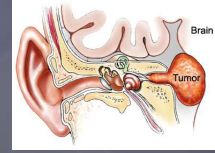
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Neurofibromatosis





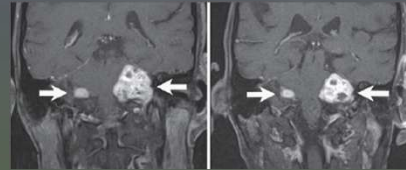
Neurocutaneous Disorders – **Neurofibromatosis 2 (Schwannomatosis)**



- Autosomal dominant, incidence 1 in 33,000
 - Half of the cases have no family hx
 - Tumor suppressor gene on chrom 22 (NF2 gene)
- Predisposes to **multiple intracranial and spinal tumors**
 - Almost all affected individuals develop **bilateral vestibular schwannomas** by age 30 years
 - May also develop schwannomas of other cranial & peripheral nerves, meningiomas, ependymomas & astrocytomas
- Average life span < 40 yrs
- Cataracts are common
- **NO lisch nodules, café-au-lait spots, axillary freckling**

Neurocutaneous Disorders – **Neurofibromatosis 2 (Schwannomatosis)** Diagnosis

- **B/L vestibular schwannomas**
or
- **FHx of NF2** (parent, sibling, or child) plus a unilateral vestibular schwannoma before age 30
or
- **Any two of the following:**
 - glioma
 - meningioma
 - schwannoma
 - juvenile posterior subcapsular/lenticular opacity (cataract) or juvenile cortical cataract



Neurocutaneous Disorders

Sturge-Weber Syndrome

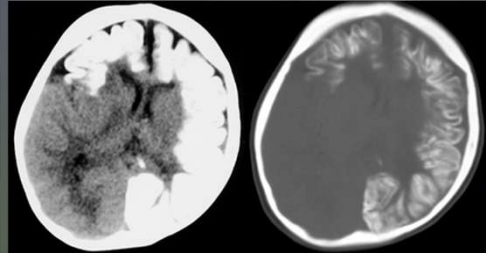
- Sporadic, not genetic
- **Angiomas of the leptomeninges**
- **Ipsilateral port-wine nevus**
- 30-50% have glaucoma
- Seizures due to ischemic injury to the brain underlying the meningeal angiomas
- Progressive ischemia can develop
 - Hemiparesis, intractable epilepsy, dementia
- Calcium is detected in gyri of brain underlying angioma → **"tram track"** on imaging
- Tx = anticonvulsants, hemispherectomy, laser surgery for facial nevus



Angiomas of the leptomeninges overlying the cerebral cortex

Ipsilateral port-wine nevus that at least covers part of the forehead and upper eyelid

Sturge-Weber Syndrome



Zoey! (2010)







Bailey is added into
the mix!
2012





2014





2016

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